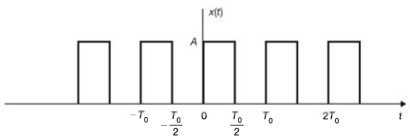
EED-201 (Signals and Systems): TUTORIAL 7

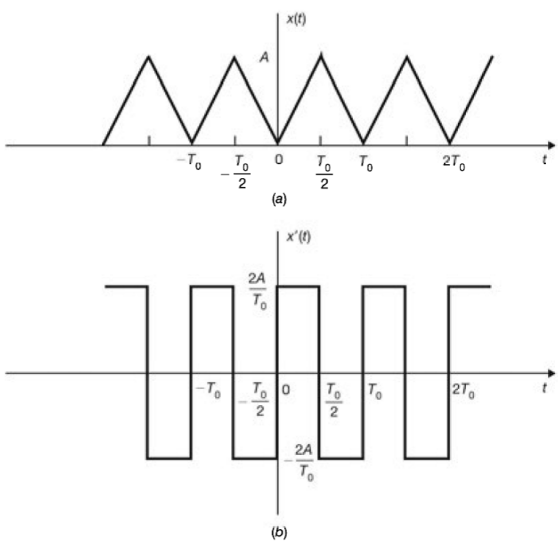
1. Determine the trigonometric fourier series of:

   i)



   ii) $\delta_{T_0}(t)$. (Note: $\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$.

2. Consider the triangular wave $x(t)$ shown in figure below. Using the differentiation technique (Use fig. (b)), find:
   i) The complex exponential Fourier series of $x(t)$
   ii) The trigonometric Fourier series of $x(t)$.

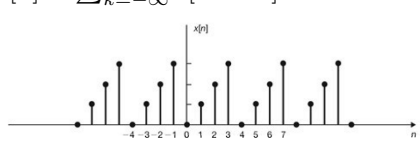


3. Let $x_1(t)$ be a continuous-time periodic signal with fundamental frequency w_1 and Fourier coefficients a_k. Given that: $x_2(t) = x_1(1-t) + x_1(t-1)$.
   How is the fundamental frequency w_2 of $x_2(t)$ related to w_1? Also, find a relationship between the Fourier series coefficients b_k of $x_2(t)$ and the coefficients a_k.

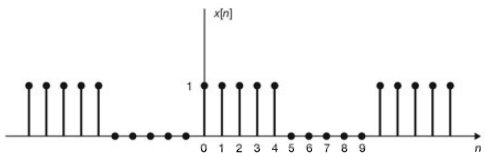
4. Let $x[n]$ be a real and odd periodic signal with period $N = 7$ and Fourier coefficients a_k. Given that: $a_{15} = j, a_{16} = 2j, a_{17} = 3j$
   Determine the values of a_0, a_{-1}, a_{-2}, and a_{-3}.

5. Determine the discrete fourier series representation of each of the following signals:
   i) $x[n] = \cos\frac{\pi}{3}n + \sin\frac{\pi}{4}n$. ii) $x[n] = \cos^2\left(\frac{\pi}{8}n\right)$.

6. Determine the fourier series coefficients of:
   i) $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k]$.
   ii)



7. Consider a periodic sequence $x[n]$ shown below. Find the fourier series coefficients c_k and sketch the magnitude spectrum $|c_k|$.



8. A discrete-time periodic signal $x[n]$ is real valued and has a fundamental period $N = 5$. The non-zero Fourier series coefficients for $x[n]$ are:
$a_0 = 1, a_2 = a_{-2}^* = e^{j\pi/4}, a_4 = a_{-4}^* = 2e^{j\pi/3}$.
Express $x[n]$ in the form: $x[n] = A_0 + \sum_{k=1}^{\infty} A_k \sin(\omega_k n + \phi_k)$.

9. Each of the two sequences $x_1[n]$ and $x_2[n]$ has a period $N = 4$, and the corresponding Fourier series coefficients are specified as: $x_1[n] \longleftrightarrow a_k, \quad x_2[n] \longleftrightarrow b_k$,
where:
$a_0 = a_3 = \frac{1}{2}a_1 = \frac{1}{2}a_2 = 1 \quad$ and $\quad b_0 = b_1 = b_2 = b_3 = 1$.
Using the multiplication property, determine the Fourier series coefficients c_k for the signal $g[n] = x_1[n]x_2[n]$.

10. When the impulse train:
$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - 4k] \tag{1}$$
is the input to a particular LTI system with frequency response $H(e^{iw})$, the output of the system is found to be:
$$y[n] = \cos\left(\frac{5\pi}{2}n + \frac{\pi}{4}\right)$$
Determine the values of $H\left(e^{jk\pi/2}\right)$ for $k = 0, 1, 2, 3$.

11. Let
$x[n] = \begin{cases} 1, & 0 \le n \le 7 \\ 0, & 8 \le n \le 9 \end{cases}$
be a periodic signal with fundamental period $N = 10$ and Fourier series coefficients a_k.
Also, let
$g[n] = x[n] - x[n-1]$
i) Determine the fourier series coefficients of $g[n]$.
ii) Using the Fourier series coefficients of $g[n]$ and the First-Difference property, determine a_k for $k \neq 0$.

12. Consider a continuous time LTI system with frequency response:
$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt = \frac{\sin(4\omega)}{\omega}$$
If the input to this system is a periodic signal:
$$x(t) = \begin{cases} 1, & 0 \le t < 4 \\ -1, & 4 \le t < 8 \end{cases}$$
with period $T = 8$, determine the corresponding system output $y(t)$.